Practice 1: Multiple choice questions:

1. **The shape attribute in NumPy or pandas returns:**
   a) The total number of missing values
   b) The number of rows and columns
   c) The column names
   d) The data types of each column

Add an example to proof your choice (with screen shot for a test code and output):

2. **Which of the following is a method and not an attribute?**
   a) shape
   b) columns
   c) dtypes
   d) mean()

Explain: _____

3. **Which command would display the first 5 rows of a DataFrame df?**
   a) df.show()
   b) df.head()
   c) df.first()
   d) df.rows(5)

Add an example to proof your choice (with screen shot for a test code and output):

4. **If df.shape returns (10, 3), it means:**
   a) 10 columns and 3 rows
   b) 3 columns and 10 rows
   c) 10 total elements
   d) 13 total cells

5. **In pandas, the skew() method is used to:**
   a) Find missing values
   b) Measure the asymmetry of data distribution

c) Sort the data
d) Remove outliers

Practice 2: Dataset World University Rankings

1. Make sure the dataset (World University Rankings.csv) is loaded using pd.read_csv() before you begin.
2. Display basic information about the dataset using built-in methods.
3. Check the shape of the dataset (i.e., number of rows and columns).
4. Display data types of each column.
5. Print the first 10 rows of the dataset only.
6. Identify and count the number of duplicate rows in the DataFrame.
7. Calculate the maximum value in the column named "**scores_industry_income**".
8. Group by location, and calculate the standard deviation of "**stats_student_staff_ratio**". *Hint: Use* `groupby()` + `std()`
9. Print the column "**scores_research**" for rows 20 to 35 only.